

Assignment 1

Group Details:

Group No: 173

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Project Details

Objective 1: Requirements Formulation

1. Select a domain and define an appropriate problem statement for an ML-based application of your choice.

1.1 Domain

Insurance

1.2 Project Title

Health Insurance Cross-Sell Prediction System

1.3 Problem Statement

An insurance company currently offers health insurance to its customers. The company wants to identify customers who are most likely to purchase vehicle insurance. Instead of contacting every customer, the marketing team wants to prioritize customers with the highest purchase probability using Machine Learning.

Business Problem

Current Challenges:

- Marketing campaigns target all customers.
- Low conversion rates.
- High customer acquisition cost.
- Inefficient resource utilization.

ML Solution

Develop an AI-powered recommendation system that predicts whether a customer will purchase vehicle insurance.

Stakeholders

Stakeholder	Goal
Marketing Team	Target interested customers
Sales Team	Increase policy sales
Customers	Receive relevant offers
Insurance Company	Improve revenue and reduce campaign costs

2. Formulate the requirement specifications and establish measurable goals for the proposed ML-based application using GR4ML concepts.

2.1 Functional Requirements

- FR1: System shall load customer information.
 - FR2: System shall preprocess the data.
 - FR3: System shall engineer additional features.
 - FR4: System shall predict purchase probability.
 - FR5: System shall display probability score.
 - FR6: System shall explain prediction using SHAP.
 - FR7: System shall allow batch prediction.
 - FR8: System shall save prediction history.
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2.2 Non-Functional Requirements

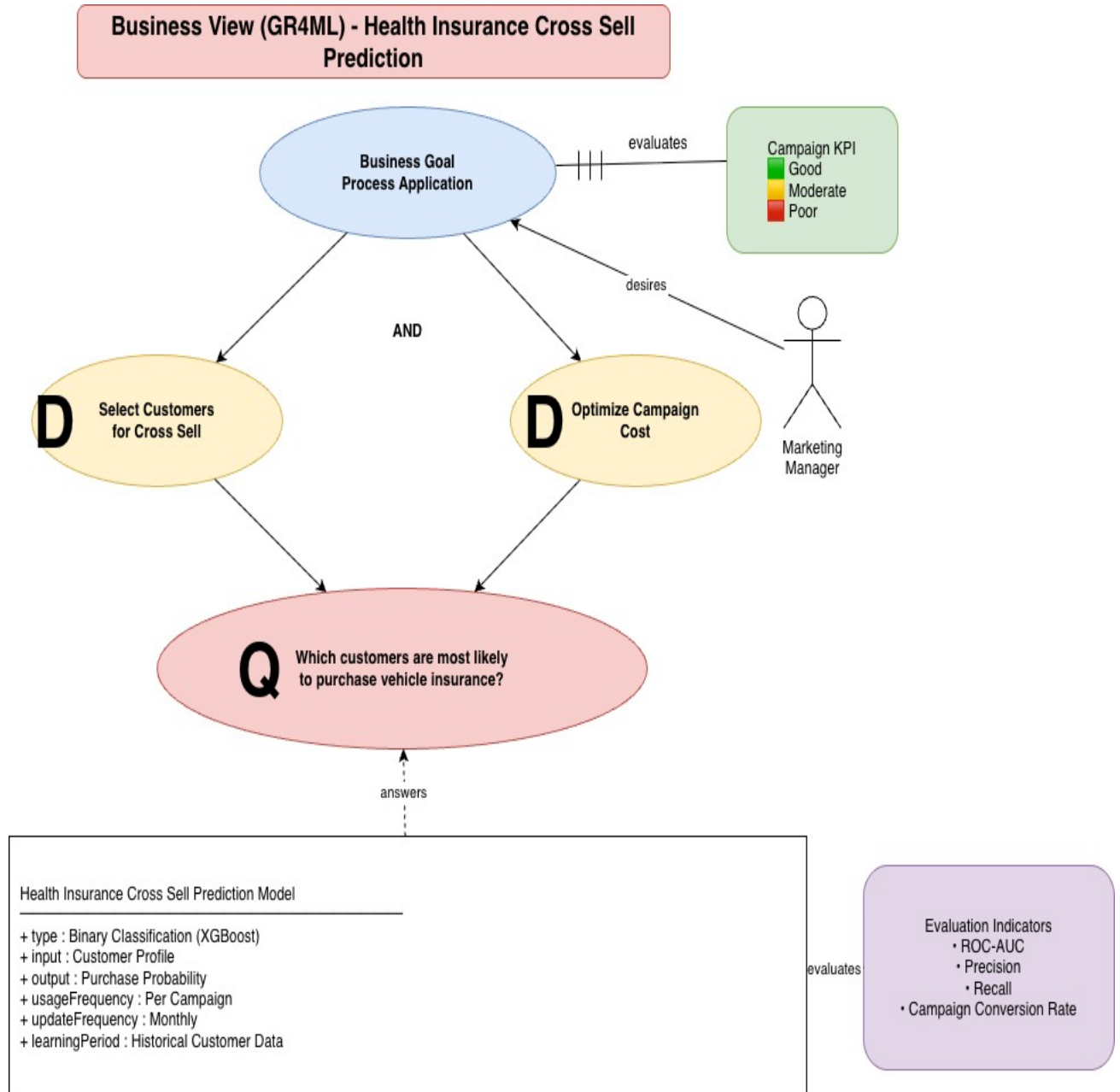
- NFR1: Prediction latency < 500 ms
 - NFR2: Availability > 99%
 - NFR3: Secure REST APIs
 - NFR4: Explainable predictions
 - NFR5: Scalable architecture
 - NFR6: Maintainability
 - NFR7: Reproducibility
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2.3 Measurable Goals

Goal	Target
ROC-AUC	>0.88
Recall	>75%
Precision	>40%
Response Time	<500 ms
Uptime	99%
Batch Prediction	10000 customers

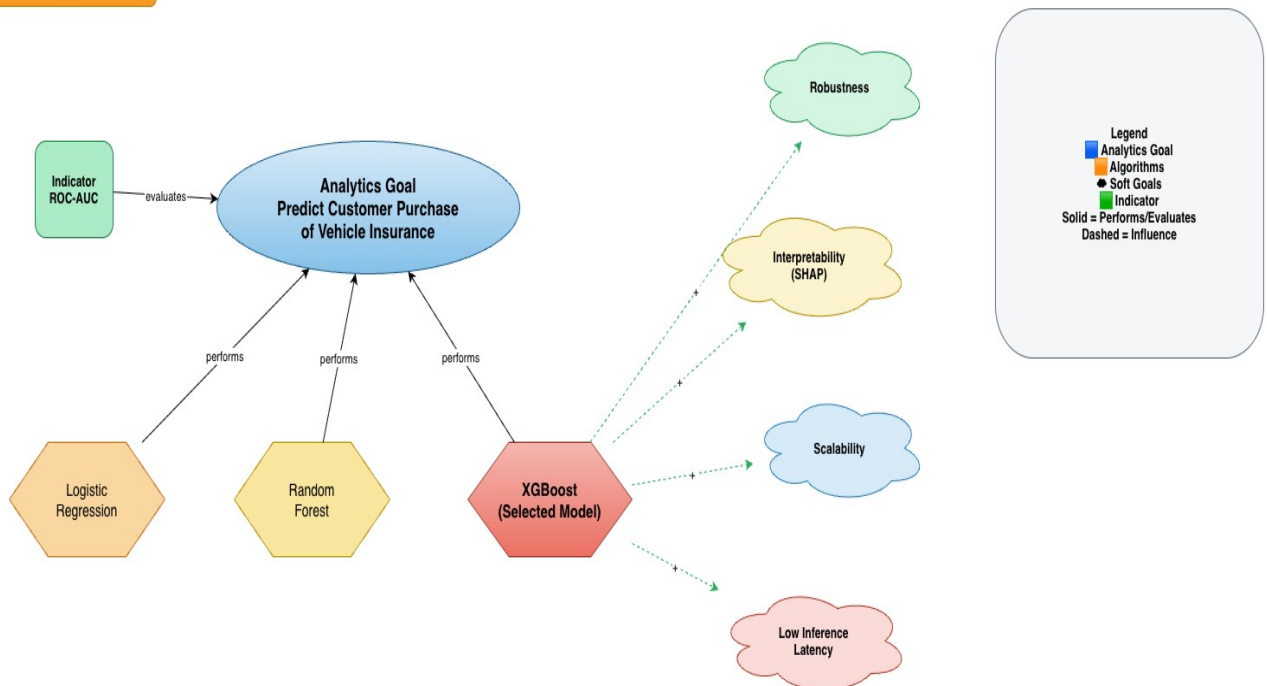
3. Using GR4ML notations, develop the following views for the application:

3.1 Business View



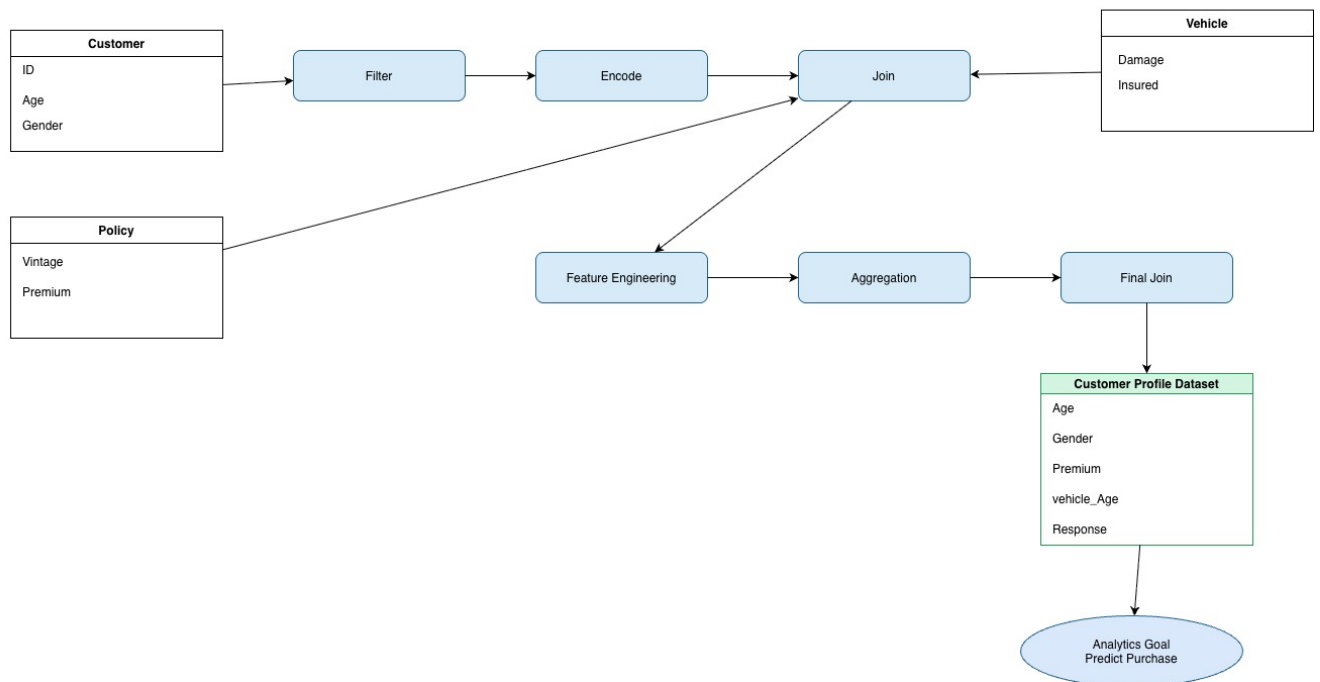
3.2 Analytics Design View

Analytics Design View



3.3 Data Preparation View

Data Preparation View



4. Identify the top three quality requirements for the application and provide justification for their selection.

4.1 Accuracy

Reason: Wrong prediction results in contacting uninterested customers, increasing operational costs.

Target: ROC-AUC > 0.88

4.2 Explainability

Reason: Insurance companies must justify recommendations and support business decisions.

Technology: SHAP

3. Scalability

Reason: The model should support predictions for millions of customers without significant performance degradation.

Technology: FastAPI + Docker + Azure

Objective 2: System Architecture

5. Draw a system architecture diagram for the selected application, clearly illustrating both ML and non-ML components. You may use any suitable tool of your choice to create the diagram.

5.1 ML Components

- Data Validation
- Feature Engineering
- XGBoost Model
- SHAP Explainability
- Prediction Service

5.2 Non-ML Components

- Streamlit UI
- FastAPI
- REST APIs
- Logging
- Model Repository
- Docker
- GitHub Actions
- Azure

